CS 184 Exam 2 Review

Selected from past CS 184/284A Exam 2s
(Summer 2025 · Spring 2025 · Spring 2024)

Today's plan

Q1 True/False warm-up (five misconceptions)

Q2 Thin lens: portrait and 1:1 macro

Q3 Russian roulette: probability $\rightarrow$ unbiasedness

Q4 Microfacet highlights: round vs. streak

Q5 The forward-Euler spiral (closer)

Q1 · True or False?

(a) Total internal reflection can only occur when light passes *into* a medium with a *higher* index of refraction.

(b) These two exposures are equivalent: $\frac{1}{200}$ s at F4, and $\frac{1}{100}$ s at F2.

(c) In a vacuum, radiance is constant along a ray — it does not diminish with distance.

(d) A microfacet BRDF with a tightly concentrated normal distribution function (NDF) appears matte.

(e) Forward (explicit) Euler can always be made stable by choosing a small enough timestep.

(f) *(bonus)* Uniform grid spatial partitions perform poorly on "teapot in a stadium" scenes.

Sources: (a) Su25 1a (b) Sp24 1e (c) Sp25 1r (d) Sp24 1r (e) Sp25 1h (f) Sp25 1t

A1 · Walkthrough (a–c)

(a) *"TIR only when entering a higher-index medium."* — **False.** It's the opposite: dense → rare (water → air), beyond the critical angle — that's when Snell's law has no transmitted solution.

(b) *"$\frac{1}{200}$ s at F4 ≡ $\frac{1}{100}$ s at F2."* — **False.** Exposure $\propto t/N^2$:

$$\frac{1/200}{16} = \frac{1}{3200} \qquad \text{vs.} \qquad \frac{1/100}{4} = \frac{1}{400}$$

The second gathers $8\times$ the light: $+3$ stops ($+2$ aperture, since area $\propto$ diameter^2; $+1$ shutter).

(c) *"Radiance is constant along a ray in vacuum."* — **True.** Distant objects look dimmer because they subtend a smaller *solid angle* — irradiance falls as $1/r^2$ while radiance per direction is unchanged.

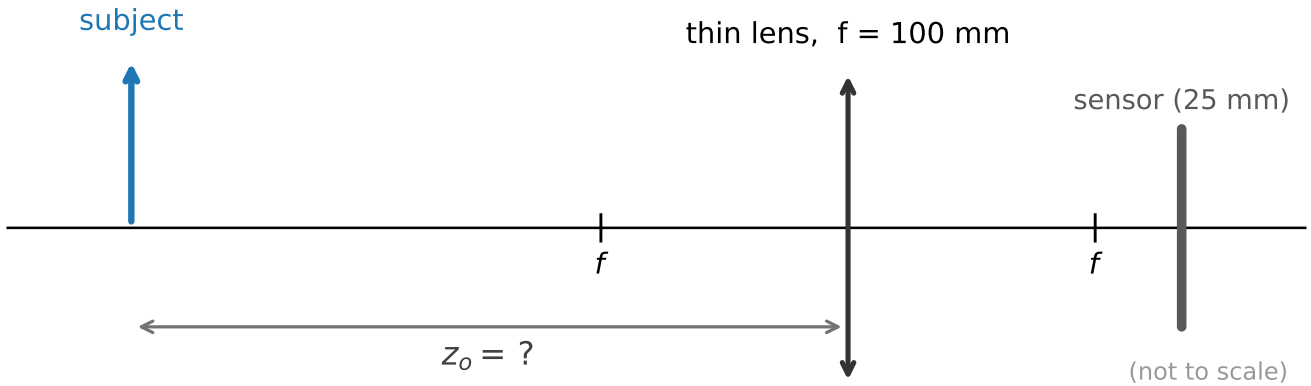
A1 · Walkthrough (d–f)

(d) *"Tightly concentrated NDF ⇒ matte."* — **False.** Concentrated = normals nearly all agree = mirror-like, tight highlights. A *broad* NDF gives matte.

(e) *"Forward Euler always stabilized by small enough Δt."* — **False** — but let's not argue it now: Q5 is a two-line *proof*. (Small steps do fix many damped systems — the "always" is what sinks it.)

(f) *"Uniform grids struggle on teapot-in-a-stadium."* — **True.** A uniform grid spends cells evenly over empty stadium space while the whole teapot lands in a few overstuffed cells. Hierarchical structures (BVH, k-d tree, octree) adapt to where geometry actually is.

Q2 · Thin lens: portrait and 1:1 macro



(i) You want exactly **1 m** of your friend's height to fill the frame. You may assume the camera is *focused at infinity*. To the nearest meter, how far away should you stand?

(ii) A shrinking ray strikes: your friend is now exactly **25 mm** tall and must still fill the frame. You may *not* assume focus at infinity. How far from the lens (in mm) should they stand?

Source: Sp24 Exam 2, Problem 3c

A2 · Walkthrough

The question: $f = 100\,\text{mm}$, $25 \times 25\,\text{mm}$ sensor. (i) 1 m of height fills the frame, focus $\approx \infty$ — distance? (ii) 25 mm friend fills the frame, no ∞ assumption — distance?

(i) Needed magnification: $m = \dfrac{25\ \text{mm}}{1000\ \text{mm}} = \dfrac{1}{40}$. Focus at infinity $\Rightarrow z_i \approx f \Rightarrow m \approx f/z_o$:

$$z_o \approx 40f = 40 \times 100\ \text{mm} = \boxed{4\ \text{m}}$$

(ii) $25\,\text{mm}$ subject on a $25\,\text{mm}$ sensor $\Rightarrow |m| = 1 \Rightarrow z_i = z_o$. Thin lens:

$$\frac{1}{z_o} + \frac{1}{z_i} = \frac{1}{f} \;\Rightarrow\; \frac{2}{z_o} = \frac{1}{100\ \text{mm}} \;\Rightarrow\; z_o = \boxed{200\ \text{mm}}\ (= 2f)$$

Q3 · Russian roulette

After every bounce, a ray **continues** with probability p_r and **terminates** with probability $1 - p_r$. Let $T =$ total bounces before termination (inclusive). Take $p_r = \frac{3}{5}$.

(a) $P(T = 1) = ?$ choices: $0,\ \frac{1}{5},\ \frac{2}{5},\ \frac{3}{5},\ \frac{4}{5},\ 1$

(b) $P(T \geq 3) = ?$ (give a closed form)

(c) $E[T] = ?$ choices: $\frac{2}{5},\ \frac{3}{5},\ 1,\ \frac{5}{3},\ 2,\ \frac{5}{2}$

(d) Does *decreasing* p_r increase, decrease, or not change rendering performance (speed)? What's the tradeoff?

Source: Su25 Exam 2, Problem 2

A3 · Walkthrough (a–d)

The question: Continue w.p. $p_r = \frac{3}{5}$, terminate w.p. $\frac{2}{5}$; T = total bounces, inclusive.

(a) Terminate right after the first bounce: $1 - p_r = \boxed{\frac{2}{5}}$

(b) Survive the first two continuation flips:
$P(T \geq 3) = p_r^2 = \boxed{\frac{9}{25}} = 1 - \left[\frac{2}{5} + \frac{3}{5} \cdot \frac{2}{5}\right]$

(c) T is geometric with success probability $1 - p_r$:
$E[T] = \dfrac{1}{1 - p_r} = \boxed{\frac{5}{2}}$

(d) **Speed increases** — paths die sooner, so each sample is cheaper. The price is **variance**: a noisier image at equal sample count. Cost traded for noise, *without bias*.

Q3 · Russian roulette — the estimator

Let C_t be the contribution of a path surviving exactly t bounces, so

$$I = \sum_{t=1}^{\infty} C_t.$$

The naive estimator truncates at M bounces over N rays,

$$I_{\text{basic}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{M} C_t,$$

which is only unbiased as $M \to \infty$.

(e) i. Using the random variable T, derive a **single-ray** Russian-roulette estimator I_{rr} of I, and **prove it is unbiased**.

(e) ii. Write the N-ray version $I_{rr,N}$.

A3 · The unbiased estimator

The question: $I = \sum_t C_t$. Build an unbiased I_{rr} from the random path length T; prove it; then the N-ray version.

Reaching bounce t means winning $t-1$ flips, so $P(T \geq t) = p_r^{t-1}$. Weight each surviving term by its reciprocal:

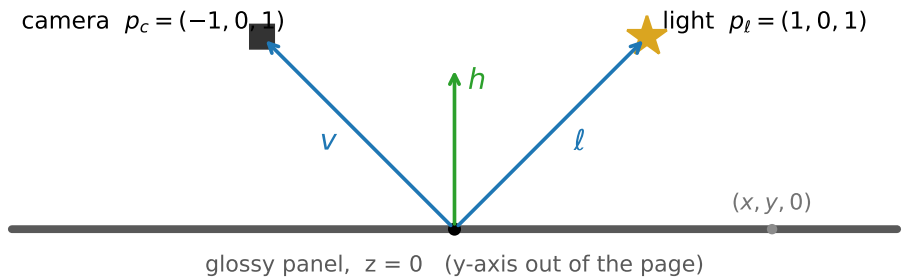
$$I_{rr} = \sum_{t=1}^{T} \frac{C_t}{p_r^{t-1}} = \sum_{t=1}^{\infty} \mathbf{1}[T \geq t] \, \frac{C_t}{p_r^{t-1}}$$

Unbiasedness — linearity plus $E[\mathbf{1}[T \geq t]] = p_r^{t-1}$:

$$E[I_{rr}] = \sum_{t=1}^{\infty} p_r^{t-1} \, \frac{C_t}{p_r^{t-1}} = \sum_{t=1}^{\infty} C_t = I \quad \blacksquare \qquad I_{rr,N} = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T_i} \frac{C_t^{(i)}}{p_r^{t-1}}$$

Q4 · Round vs. streaky highlights — the geometry

Glossy panel in $z = 0$; point light at $p_\ell = (1, 0, 1)$; camera at $p_c = (-1, 0, 1)$.



(i) For each surface point $(x, y, 0)$, write unit vectors $\ell(x, y)$ and $v(x, y)$ toward the light and the camera.

(ii) A microfacet reflects light to the camera when its normal is parallel to the half vector. Compute $h(x, y) = \dfrac{\ell + v}{\|\ell + v\|}$.

Source: Sp25 Exam 2, Problem 7b

A4 · Walkthrough (i–ii)

The question: Panel $z = 0$, light $(1, 0, 1)$, camera $(-1, 0, 1)$: find ℓ, v, then $h = \frac{\ell+v}{\|\ell+v\|}$.

(i) Offsets $p_\ell - (x, y, 0)$ and $p_c - (x, y, 0)$, normalized:

$$\ell(x, y) = \frac{(1 - x,\ -y,\ 1)}{\|(1 - x,\ -y,\ 1)\|} \qquad v(x, y) = \frac{(-1 - x,\ -y,\ 1)}{\|(-1 - x,\ -y,\ 1)\|}$$

(ii) Near the origin the two denominators nearly match and the ± 1 x-components nearly cancel, so $\ell + v$ tilts opposite (x, y); for small $|x|, |y|$:

$$h(x, y) \ \approx\ \frac{(-x,\ -y,\ 1)}{\sqrt{x^2 + y^2 + 1}}$$

Q4 · Round vs. streaky highlights — the materials

Microfacet NDFs (unit normal N, upper hemisphere via $H(N_z)$); $\sigma_y \gg \sigma_x$ models a surface combed along y:

$$D_{\text{iso}}(N) = \frac{H(N_z)}{2\pi\sigma^2}\exp\left[-\frac{N_x^2+N_y^2}{2\sigma^2}\right] \quad D_{\text{aniso}}(N) = \frac{H(N_z)}{2\pi\sigma_x\sigma_y}\exp\left[-\frac{N_x^2}{2\sigma_x^2}-\frac{N_y^2}{2\sigma_y^2}\right]$$

Take as given $\quad h(x,y) \approx \dfrac{(-x,\,-y,\,1)}{\sqrt{x^2+y^2+1}}.$

(iii) Write the reflected radiance toward the camera: $L_{\text{iso}}(x,y) \propto D_{\text{iso}}(h)$ and $L_{\text{aniso}}(x,y) \propto D_{\text{aniso}}(h)$.

(iv) Round or linear highlight for each? Justify briefly.

A4 · Walkthrough (iii–iv)

The question: Plug $h \approx (-x, -y, 1)/\sqrt{x^2+y^2+1}$ into D_{iso}, D_{aniso} — what shape is each highlight?

With $N = h$: $\quad N_x^2 + N_y^2 = \dfrac{x^2+y^2}{x^2+y^2+1}$, so

$$L_{\text{iso}}(x,y) \propto \exp\left[-\frac{x^2+y^2}{2\sigma^2(x^2+y^2+1)}\right] \approx \exp\left[-\frac{x^2+y^2}{2\sigma^2}\right]$$

$$L_{\text{aniso}}(x,y) \propto \exp\left[-\frac{x^2}{2\sigma_x^2(x^2+y^2+1)} - \frac{y^2}{2\sigma_y^2(x^2+y^2+1)}\right]$$

$$\approx \exp\left[-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right]$$

(iv) The highlight *is* a contour map of the NDF. Isotropic: level sets $x^2+y^2 = \text{const}$ are circles $\Rightarrow$ **round**. Anisotropic, $\sigma_y \gg \sigma_x$: ellipses hugely elongated along $y \Rightarrow$ **linear streak**. *(T/F (d) settled: tighter NDF = shinier, not matte.)*

Q5 · The forward-Euler spiral

Particle starts at $p^0 = (1, 0)$ with position-dependent velocity

$$v(p) = (p_y,\, -p_x),$$

always tangent to the circle of radius $\|p\|$ — the exact trajectory is the unit circle.

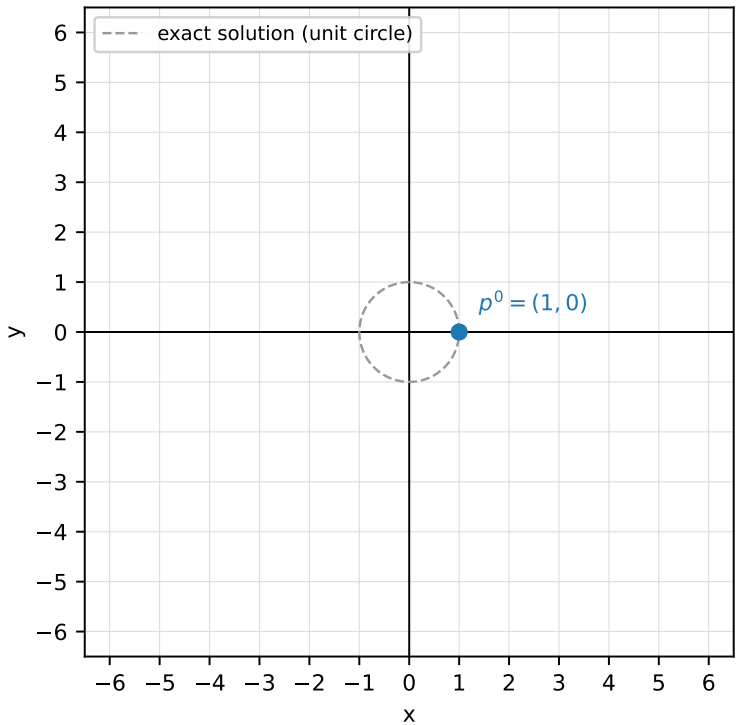
Forward Euler, $\Delta t = 1$:

$$p^{n+1} = p^n + \Delta t\, v(p^n)$$

Compute and plot p^1 through p^5.

Then: qualitatively, what changes with $\Delta t = 0.1$?

Source: Sp24 Exam 2, Problem 6a.i



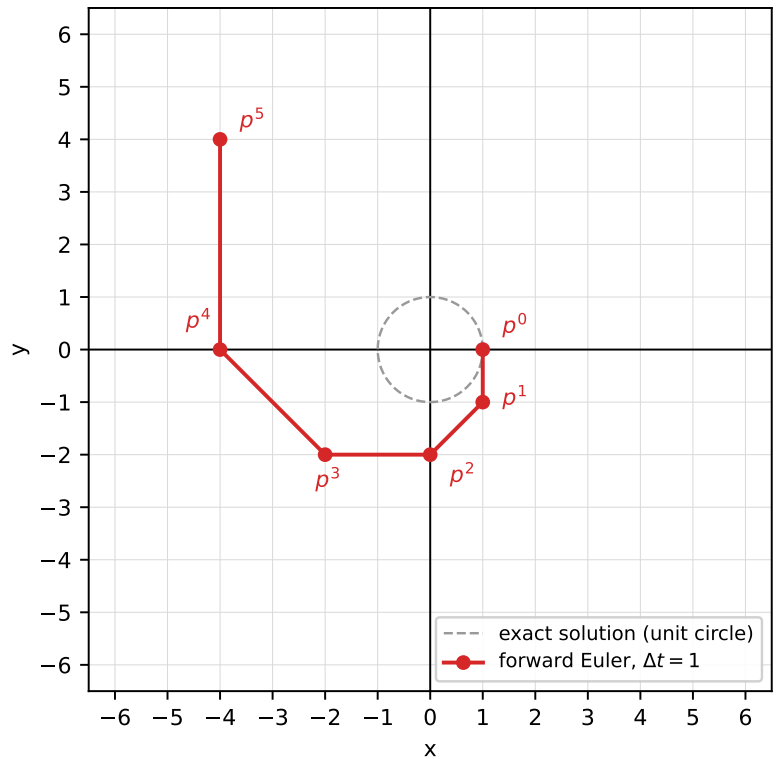
A5 · Five steps

The question: $p^0 = (1, 0)$, $v(p) = (p_y, -p_x)$, $\Delta t = 1$: five Euler steps $p^{n+1} = p^n + \Delta t\, v(p^n)$.

- $p^0 = (1, 0), \;\; v = (0, -1)$
- $p^1 = (1, -1), \;\; v = (-1, -1)$
- $p^2 = (0, -2), \;\; v = (-2, 0)$
- $p^3 = (-2, -2), \;\; v = (-2, 2)$
- $p^4 = (-4, 0), \;\; v = (0, 4)$
- $p^5 = (-4, 4)$

Radii: $1,\ \sqrt{2},\ 2,\ 2\sqrt{2},\ 4,\ 4\sqrt{2}$

— doubling every two steps.



A5 · Why it must spiral

The question: Exact orbit is the unit circle. What does Euler do for *any* Δt — and for $\Delta t = 0.1$?

$v(p) \perp p$ and $\|v\| = \|p\|$, so by Pythagoras:

$$\|p^{n+1}\|^2 = \|p^n + \Delta t\, v(p^n)\|^2 = \|p^n\|^2 + \Delta t^2 \|p^n\|^2 = (1+\Delta t^2)\, \|p^n\|^2$$

Every step multiplies the "energy" by $1 + \Delta t^2 > 1$ — for *any* $\Delta t > 0$.

With $\Delta t = 0.1$: the path hugs the circle but still drifts outward ($\approx 0.5\%$ radius growth per step). Shrinking the step *slows the leak; it never stops it*.

So T/F (e) is **False**: no timestep makes forward Euler stable on this undamped rotation — which is why implicit Euler, symplectic integrators, and Verlet exist.

Originals & official answer keys

Summer 2025 Exam 2
https://cs184.eecs.berkeley.edu/assets/su25/su25exam2.pdf
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